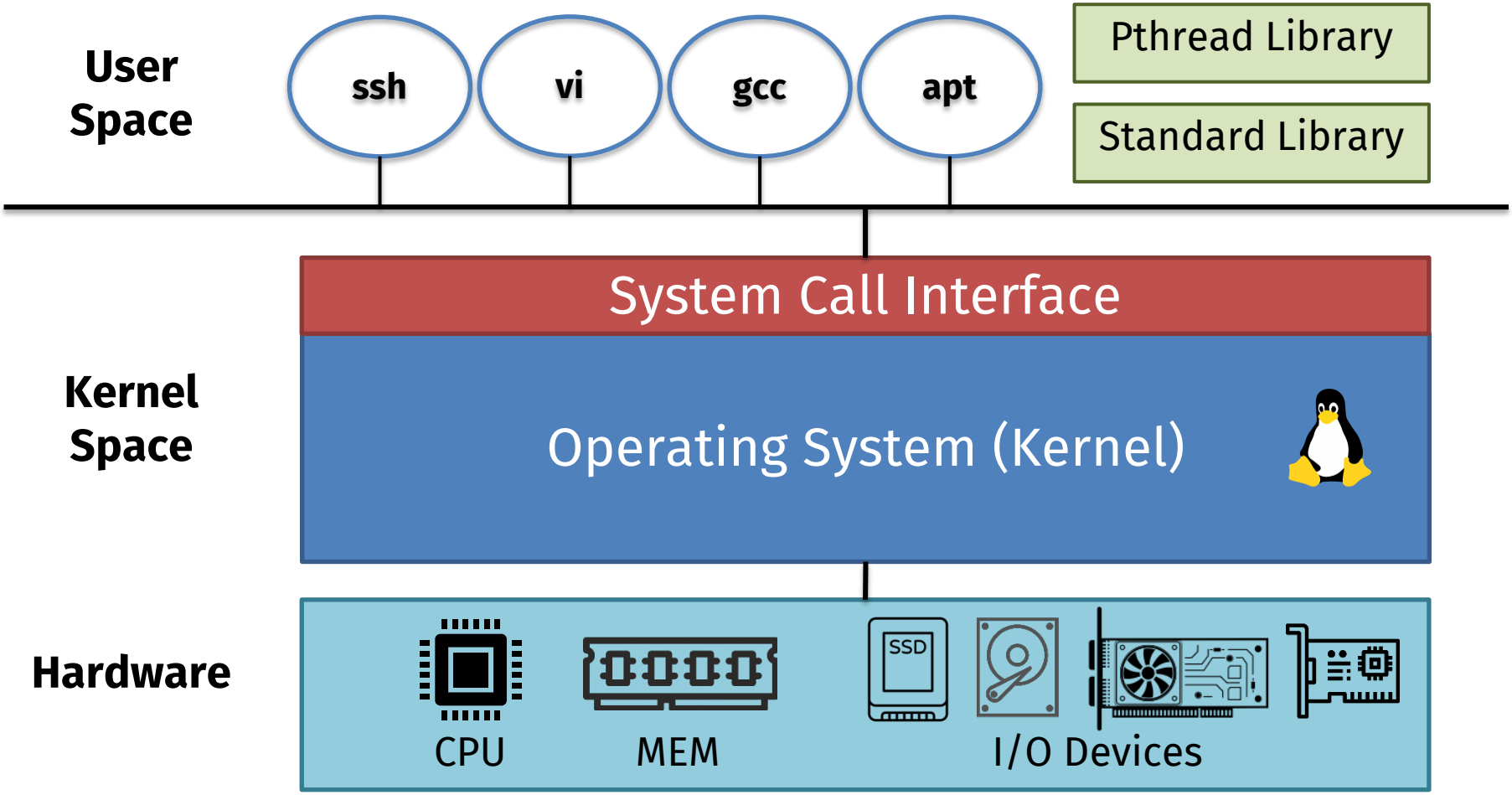
# SWE2024
## System Programming Lab

SKK UNIVERSITY

# Course Information

- **Schedule**
  - 18:00 – 21:45 (Wed): #26312
- **TA**
  - Luke Albano (lukealbano@arcs.skku.edu)
  - Jongyoon Kim (jongyoon.kim@skku.edu)
- **Use iCampus**
  - Announcements, lecture contents
  - Weekly lab assignment submission
  - Programming assignment submission
  - Q&A board (문의게시판)

# Course Outline (1)



# Course Outline (2)

| | |
|---|---|
| **Course Description:** | This course is intended to make students be familiar with Linux systems. We will learn how to install and setup your own Linux system and review the basic Linux commands. We move on to various system calls provided by Linux systems for advanced programming. No prior knowledge on the Linux system is required. |

- **Why we use Linux?**
  - Used in many scientific and industrial settings
  - Internet servers and services run on Linux
- **How to use Linux?**
  - Get used to programming environment in Linux
- **How to write advanced programs on Linux?**
  - Learn various system calls provided by Linux

# Course Outline (3)

- **Very basic Linux commands**
  - Shell, text editor, compiler
- **Basic Linux system calls**
  - File I/O, Process management
  - Inter-Process Communication (IPC)
- **Network programming**
  - Sockets, I/O Multiplexing
- **Concurrent programming**
  - Processes, Threads, Synchronization

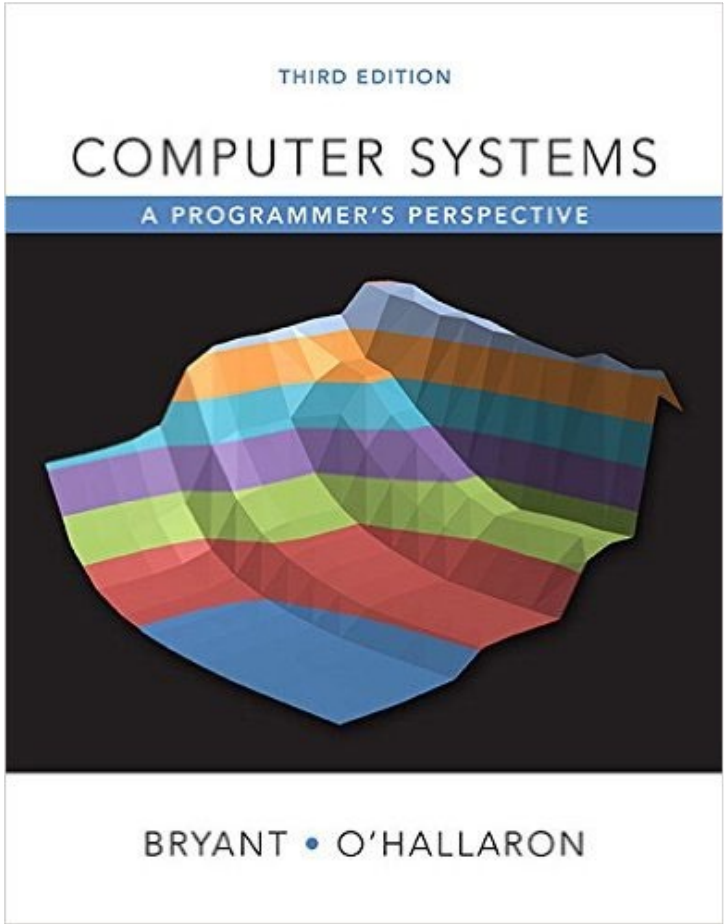
# Reference

- **Computer Systems: A Programmer's Perspective**
  - Randal E. Bryant and David R. O'Hallaron,

    Third Edition,

    Pearson Education Inc. 2015.

  - http://csapp.cs.cmu.edu



# Grade

- **Grading Policy**
  - Class attendance (10%) + Lab exercise (15%)
  - 3 Programming assignments (10+15+20 → 45%)
  - Final Exam (30%)

- **Attendance Policy**
  - Up to two absences are allowed without it affecting the final grade.
  - Four absences result in an automatic **F**.

# Academic Misconduct

- **What is cheating (plagiarism)?**
  - Submitting other's works (including ones from internet)
  - Allowing others to use your work for their assignments
- **What is NOT cheating?**
  - Helping others use systems or tools
  - Discussing with others on high-level design issues
  - Helping others debug their code
- **Penalty for cheating:**
  - Penalty on grade (both grades for original copy and its copies)
  - Severe misconduct will be reported to university
- **Ask helps to us if you experience any difficulty!**

# Course Schedule (1)

| Week | Theme | Contents |
|---|---|---|
| 1 | Introduction | • Linux install with VirtualBox, text editor (Vim) |
| 2 | Programming in Linux (1) | • Compilling, debugging codes in Linux environment |
| 3 | Programming in Linux (2) | • Git usage, writing makefile & shell scripts |
| 4 | File I/O | • Basic file I/O system calls (open, read, write, lseek) |
| 5 | Process | • Creating, managing process |
| 6 | Signal | • Sending / handling signals |
| 7 | IPC | • IPC mechanisms (pipe, redirection, ...) |
| 8 | No class | • N/A |

# Course Schedule (2)

| Week | Theme | Contents |
|---|---|---|
| 9 | System V IPC | • Shared memory, named pipe, message queue, … |
| 10 | Socket | • Basic client/server program using socket |
| 11 | Pthread | • pthread API usage |
| 12 | Concurrent Programming | • Multi-process, multi-thread programming<br>• Multiplexing I/O |
| 13 | Synchronization | • Synchronization with spinlocks, mutex and conditional vars |
| 14 | Daemon | • Linux daemon programming |
| 15 | No class | • N/A |
| 16 | Final Exam | |

# Setting Up an Ubuntu VM
## (or Installing Ubuntu as your OS)

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# Steps

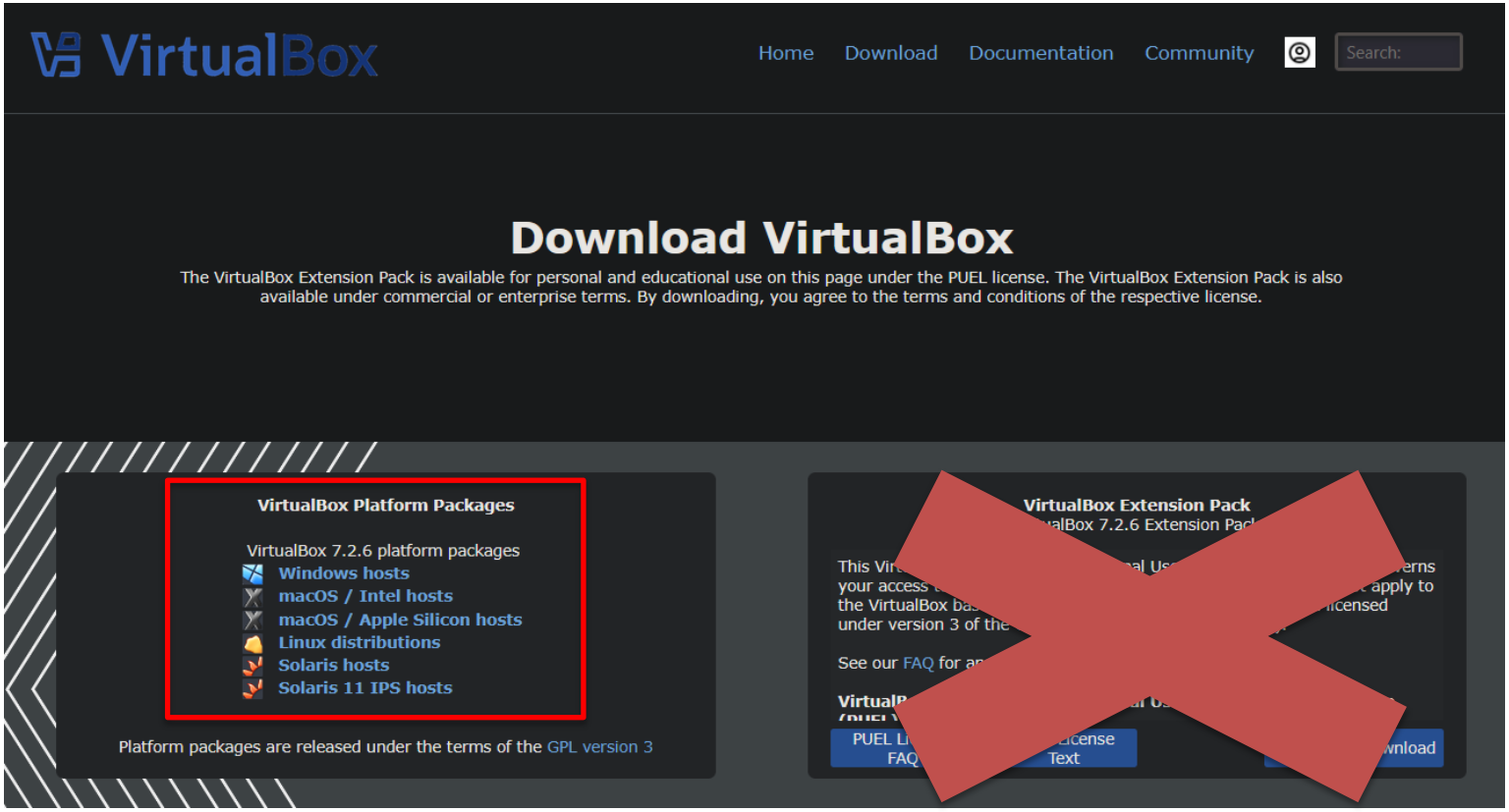
1. Install VirtualBox on your computer

2. Create a virtual machine (VM)

3. Setting up a virtual machine (VM)

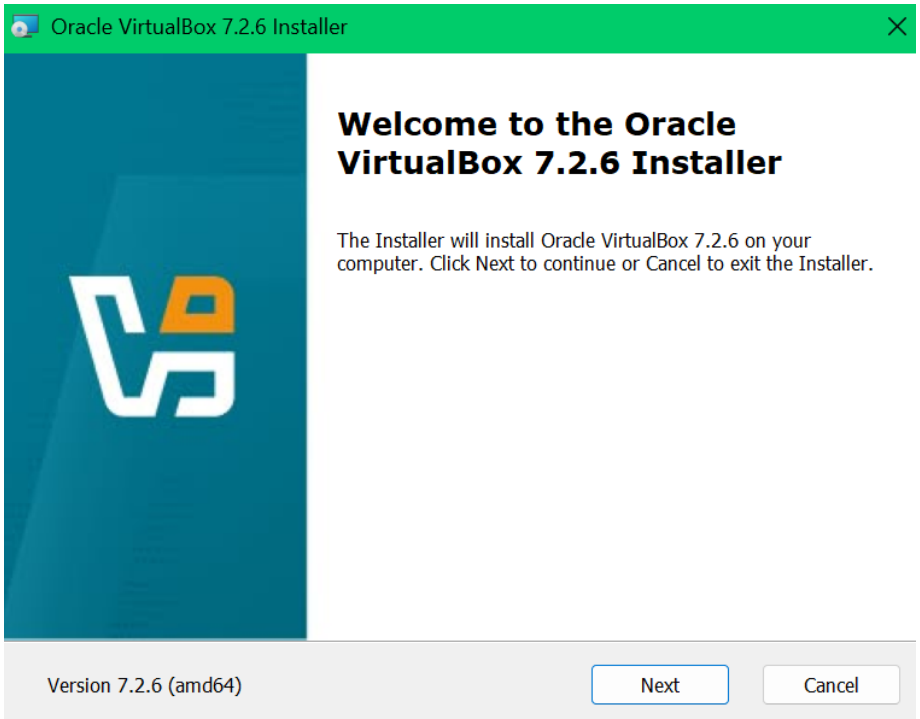
# Installing VirtualBox (1)

- **Go to VirtualBox website**
  - https://www.virtualbox.org/
- **Download installation binary**



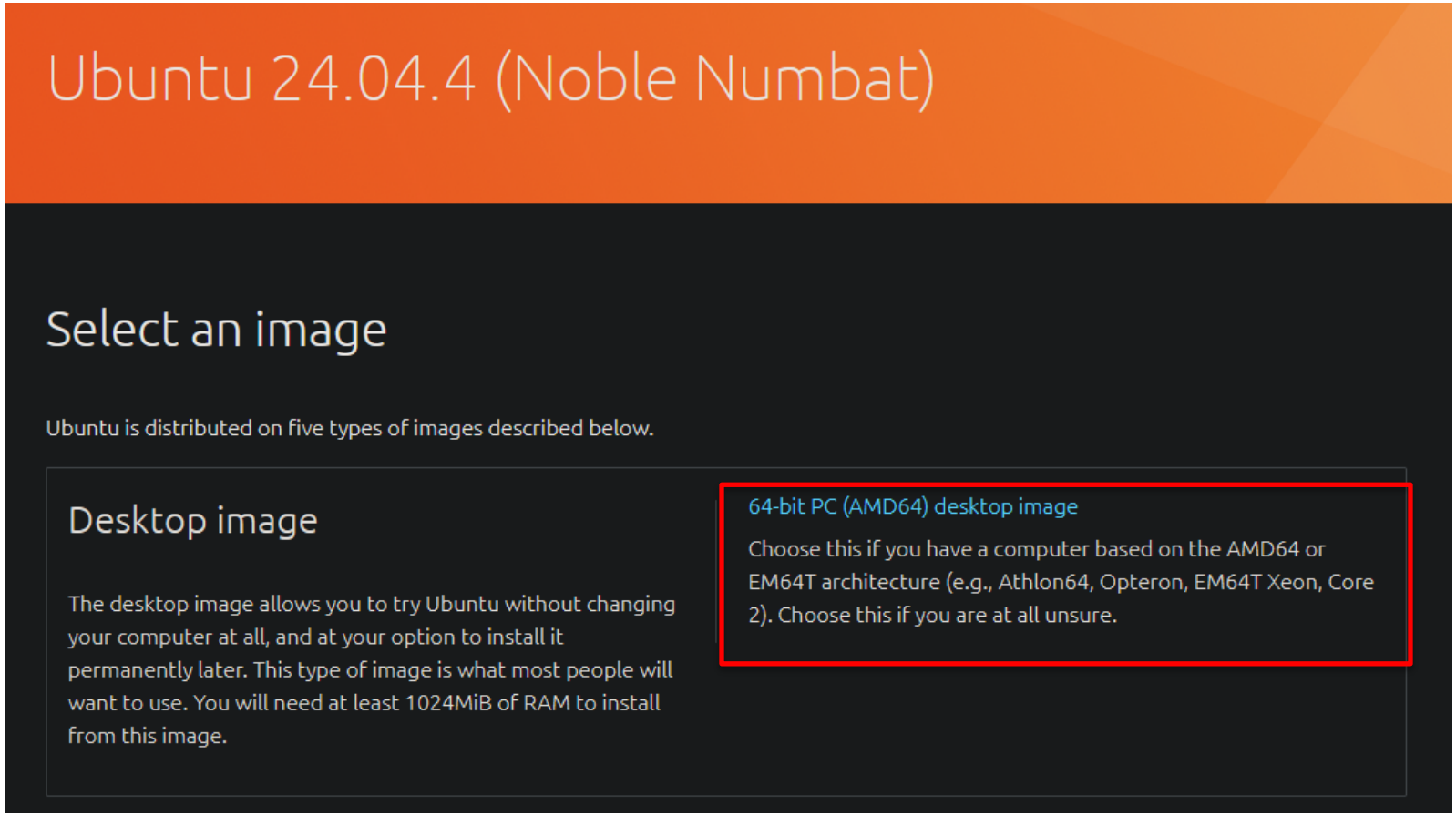
# Installing VirtualBox (2)

- Next, Next, Next, Yes, Yes, Next, Install, Finish!
- Note: You might have to restart your computer before proceeding after installing VirtualBox.



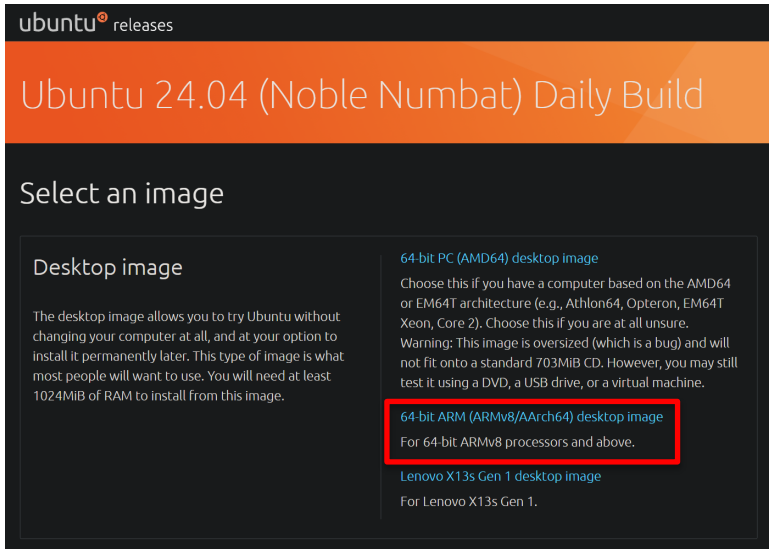
# Downloading Ubuntu

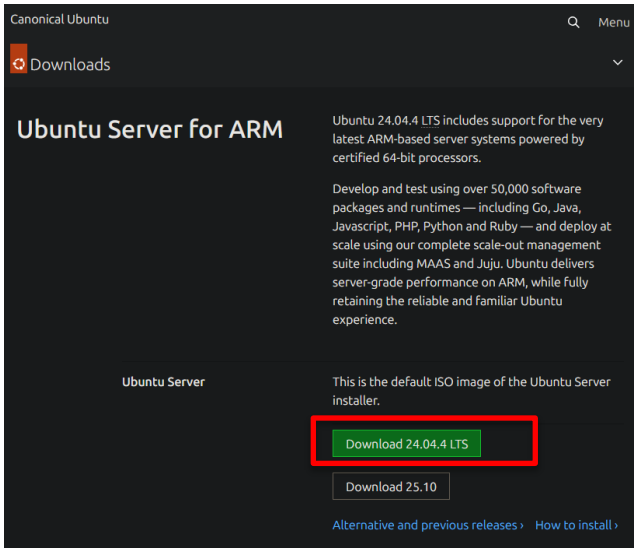
- Go to https://releases.ubuntu.com/noble/
- Download the desktop image (6.2GB)



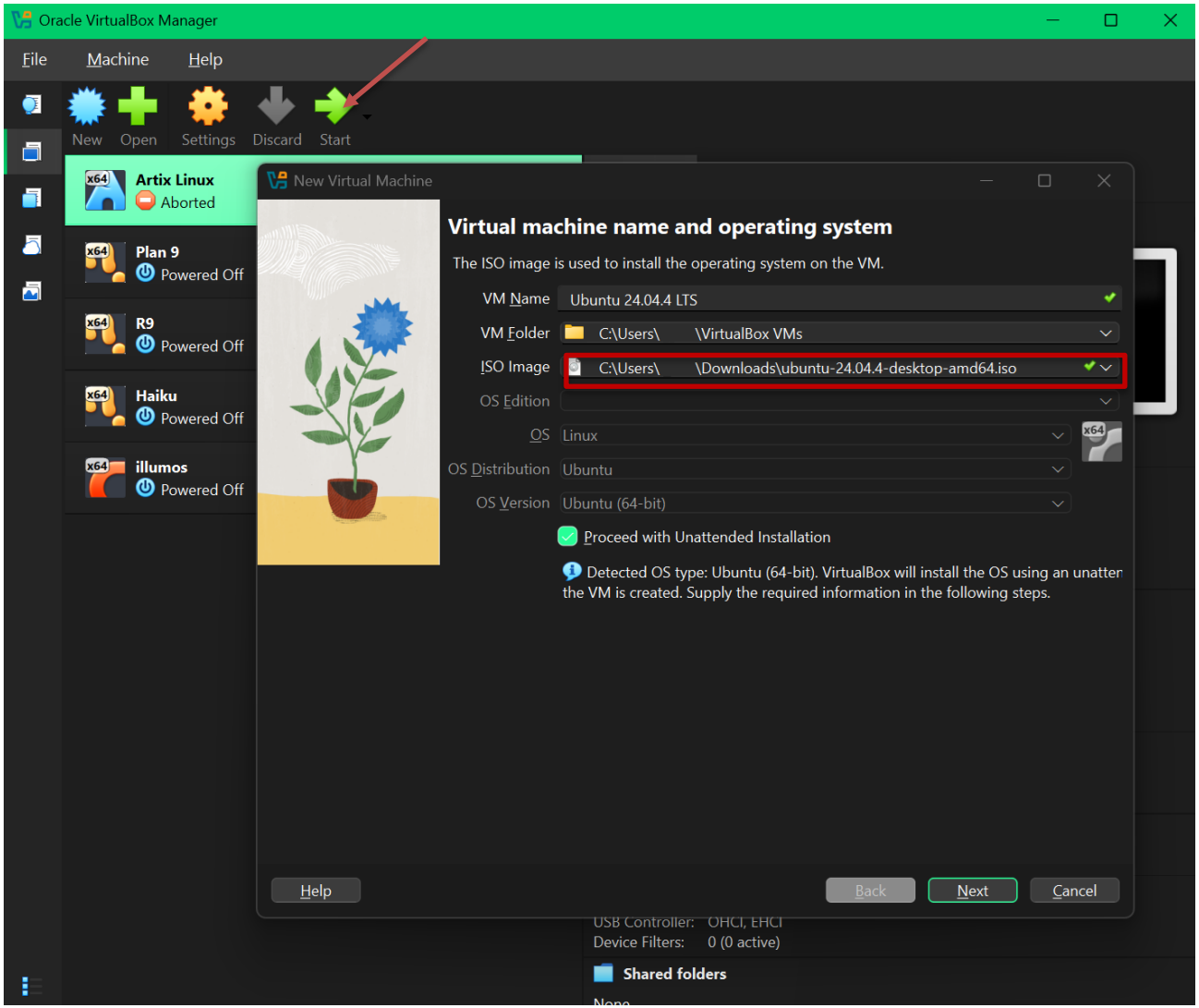
# Downloading Ubuntu (M1~M4)

- **If you are using an ARM-based computer like Apple Silicon Macs, you cannot use the previous image.**
- **You have two choices for the image:**
  - Ubuntu Desktop (has graphical interface, but might be unstable):
    - https://cdimage.ubuntu.com/noble/daily-live/current/
    - After installing Ubuntu, you must run `sudo apt update; sudo apt upgrade; sudo apt install build-essential`.
  - Ubuntu Server (no graphical interface but stable):
    - https://ubuntu.com/download/server/arm

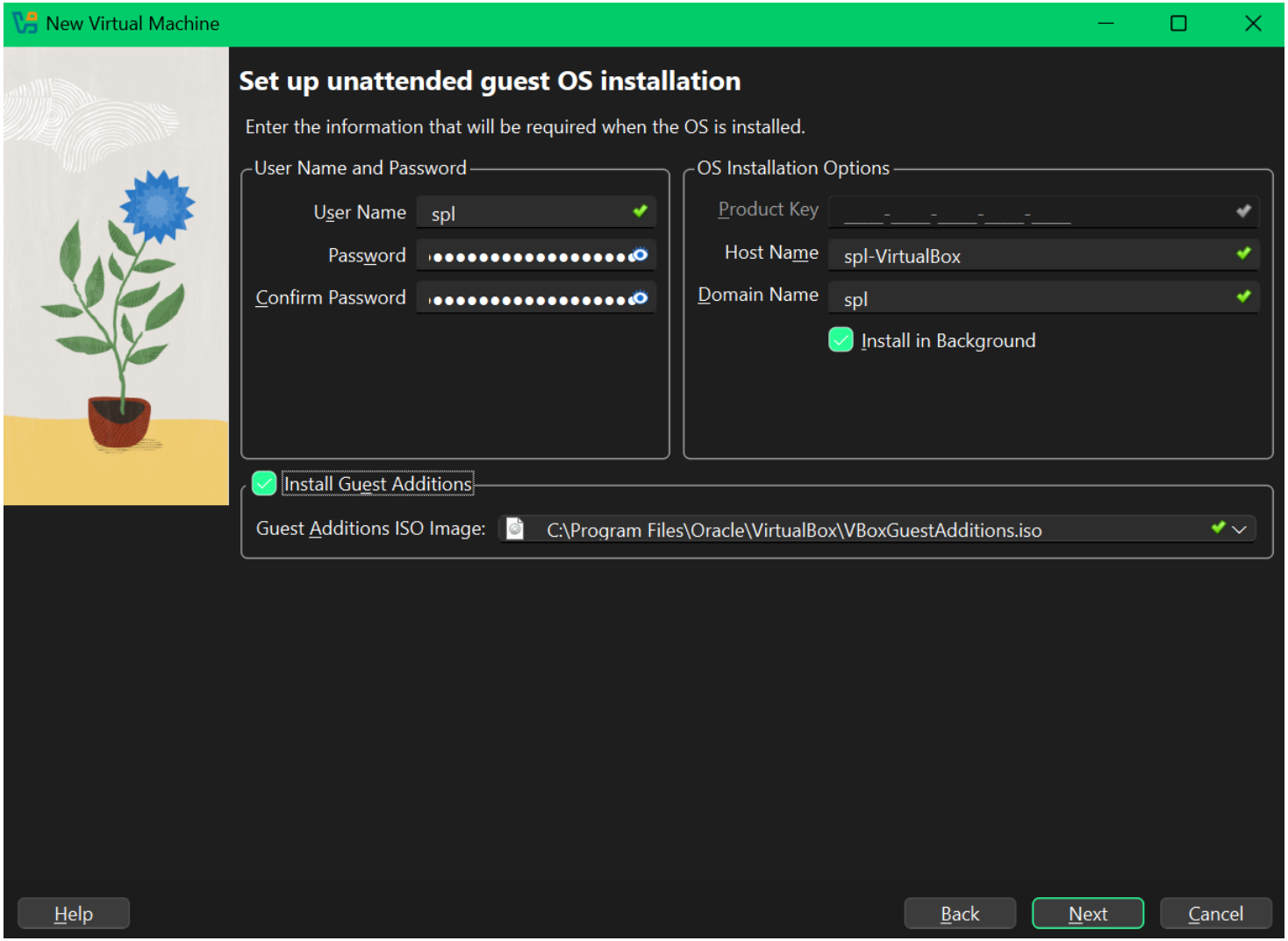


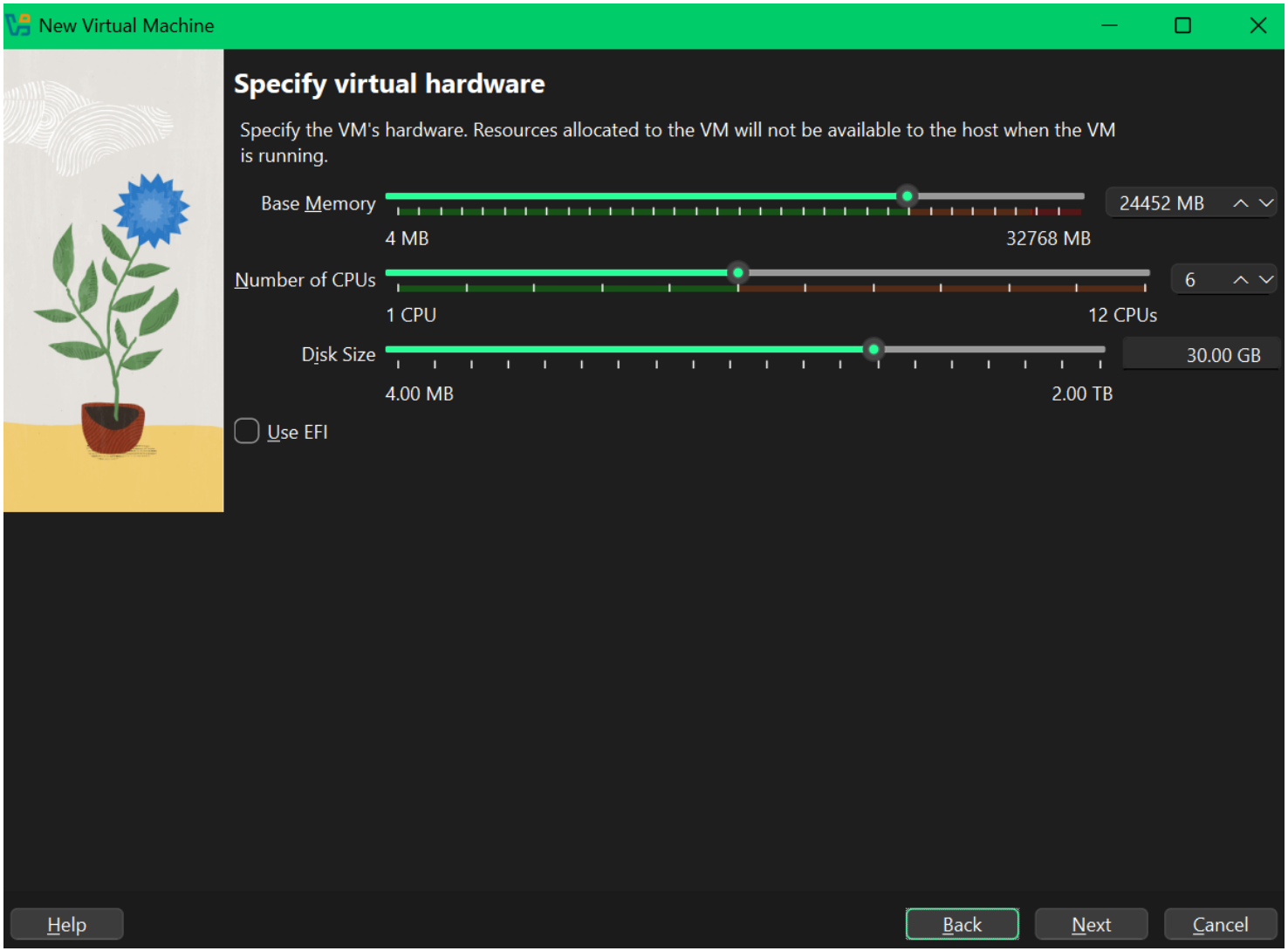
# Creating a VM (1)



# Creating a VM (2)



# Creating a VM (3)

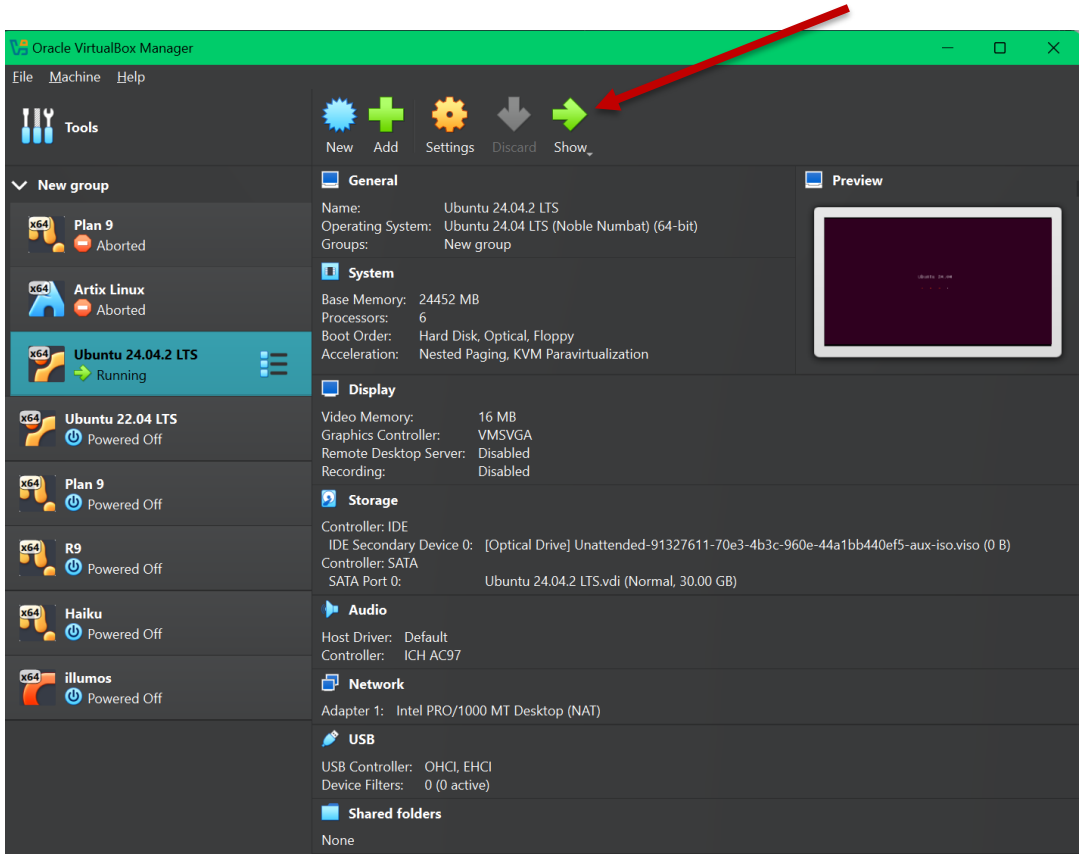


# Creating a VM (5)

- **After clicking “Finish”, Ubuntu will be installed automatically.**
- **Installation might take several minutes.**
- **Installation can sometimes fail.**
  - Try installing Ubuntu again
  - If you encounter “vmwgfx seems to be running on an unsupported hypervisor.”, try changing the graphics controller to VBoxVGA or VBoxSVGA.

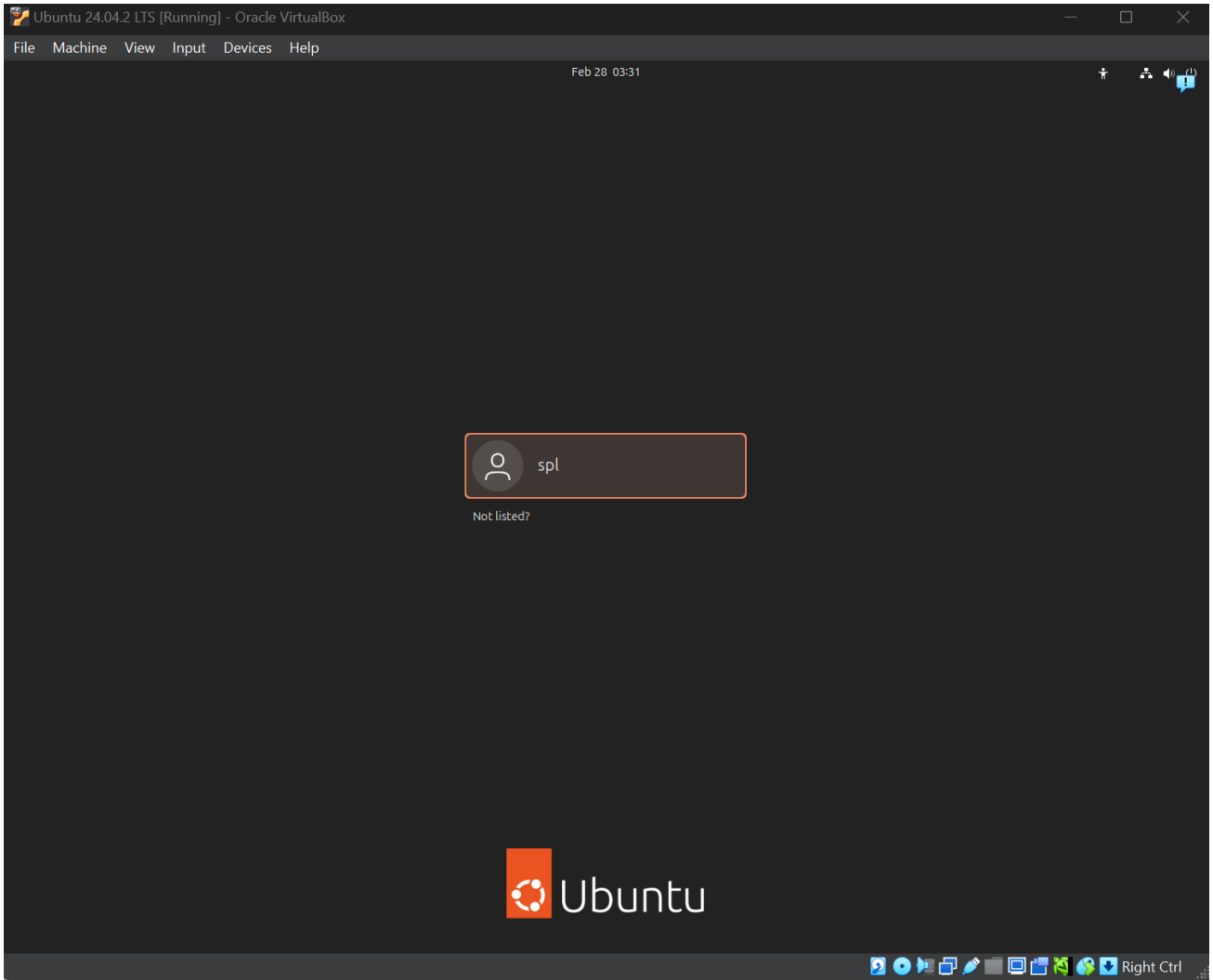
# Starting a VM (1)

- After the OS has been installed, double-click the VM name on the side menu or click the *Show* button.



# Starting a VM (2)

- You will be greeted with this window.



# Starting a VM (3)

- **Let's test your VM**
- **Press *Ctrl + Alt + T* to launch a terminal (shell)**
- **Type the following commands:**

```
$ sudo apt update
$ sudo apt upgrade
$ sudo apt install build-essential
```

# Coding in Linux

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# Text editor – Vim (1)

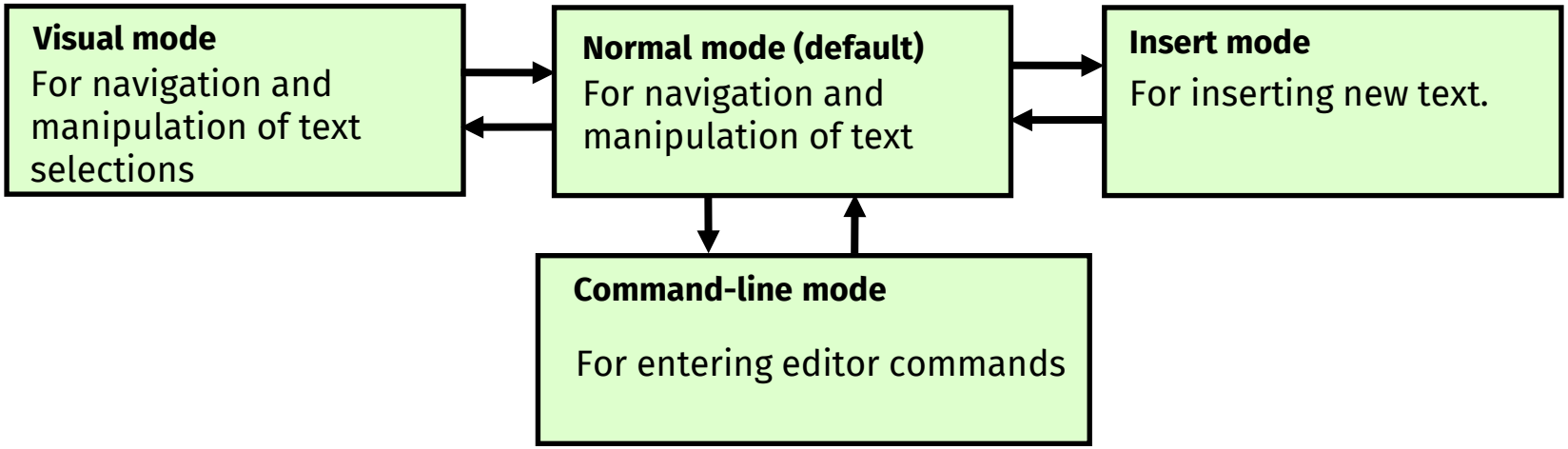
- **Vi & Vim**
  - Vi is the default editor in all UNIX operating systems
  - **It may be hard to learn, but it is useful**
  - Vi in Linux is usually Vim (Vi Improved)
  - You can easily install Vim with command:

```
$ sudo apt install vim
```

Note: $ means we are using the command-line, do not copy it.

# Text editor – Vim (2)

- **Vim has 6 modes**
  - Normal mode
  - Insert mode
  - Visual mode & select mode
  - Command-line mode & Ex-mode
- **You can easily change between modes.**

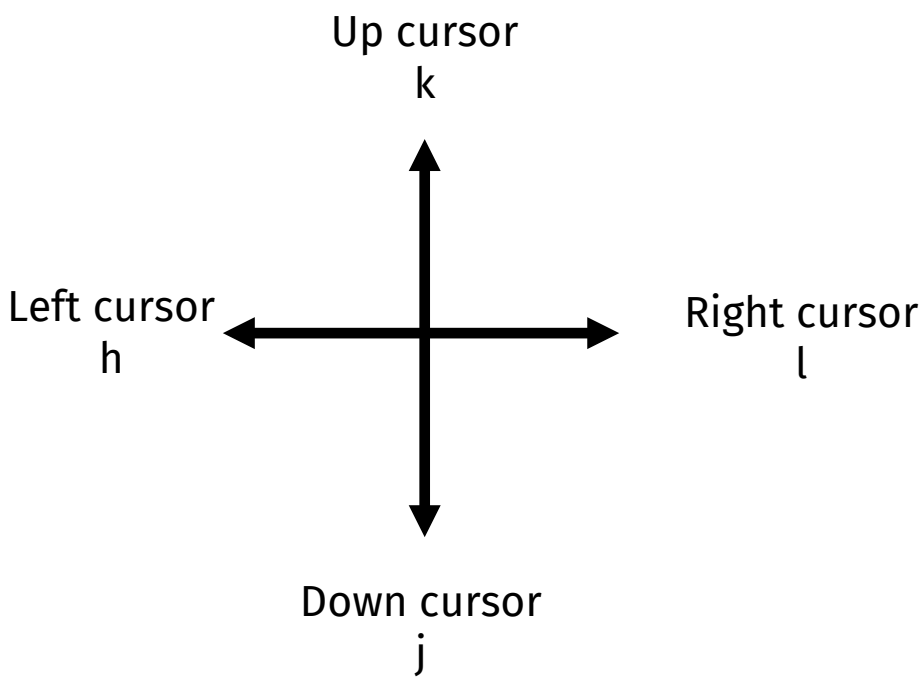


# Text editor – Vim (3)

- **Basic interface**
  - **i**, **a**, **o**, **s** : Insert mode
  - **v** : Visual mode
  - **h**, **j**, **k**, **l** or ( cursor keys ) : Cursor movement
  - ‘**:**’ ‘**/**’ : Command mode
  - ‘**Esc**’ : Return to normal mode

# Text editor – Vim (4)

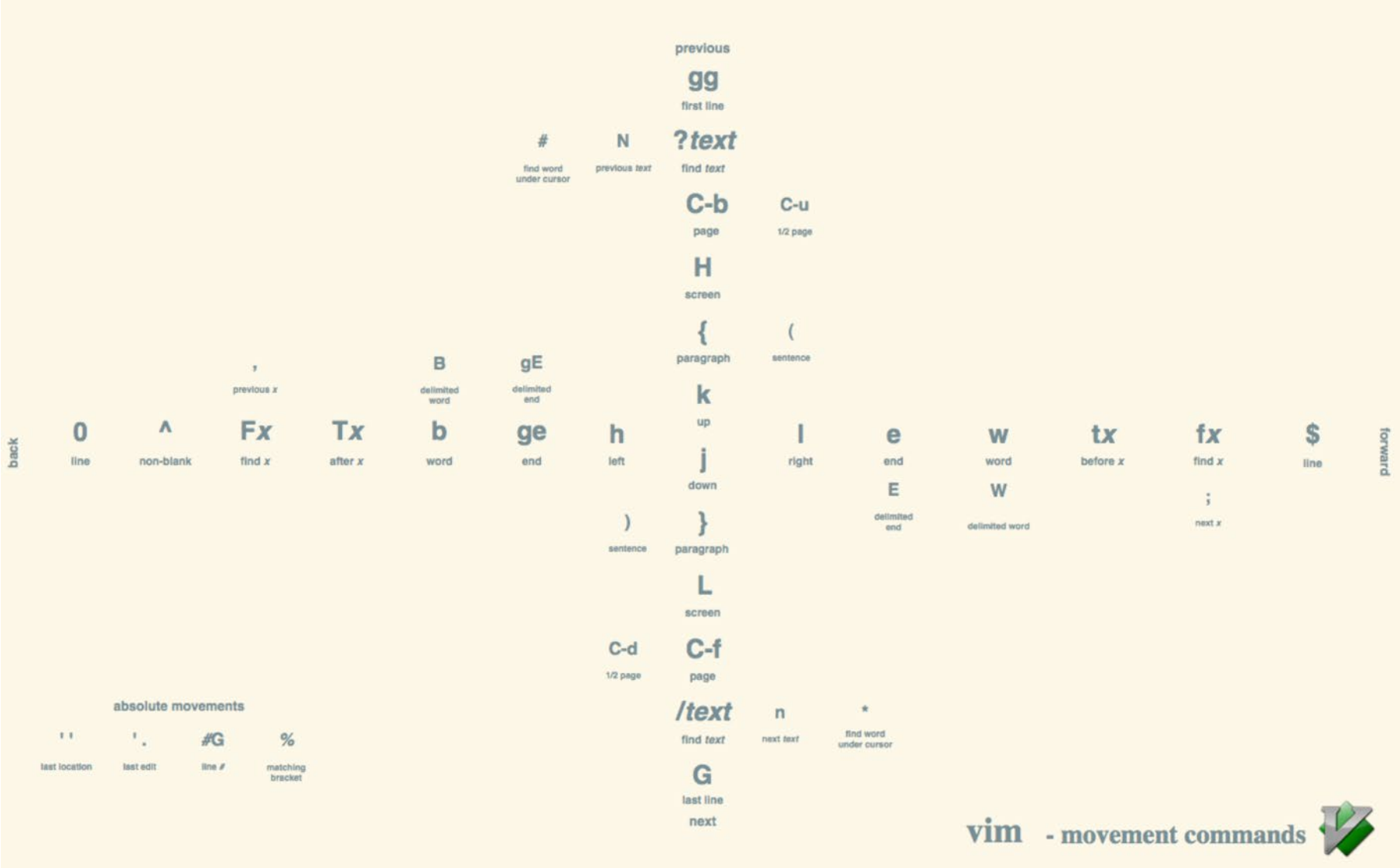
- **Cursor movement in normal mode**



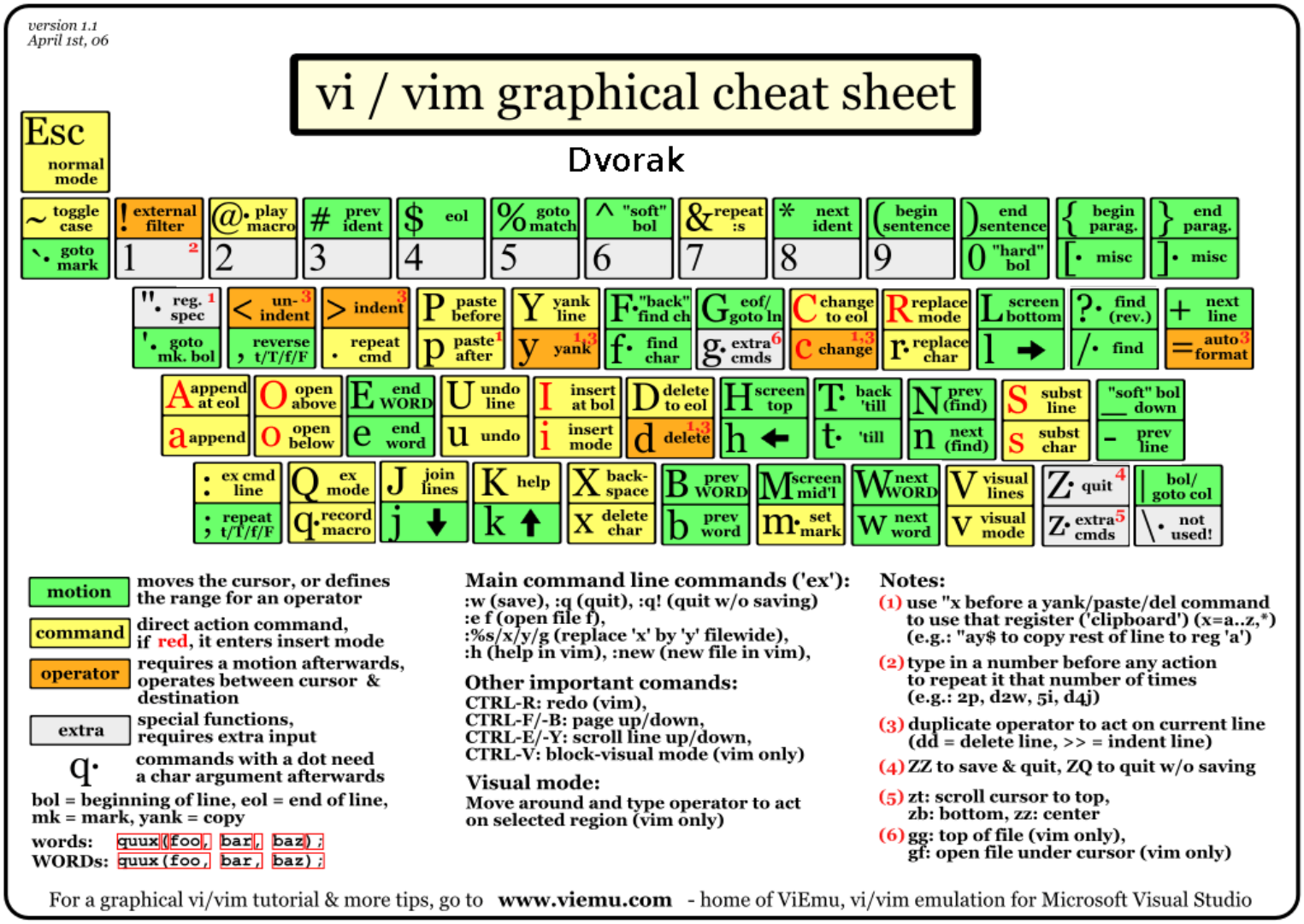
# Text editor – Vim (5)

- **Exiting Vim in command-line mode**
  - To save file
    - `:w (name)`
    - **You can omit the name if the file exists.**
  - To quit without saving
    - `:q`
  - To forcefully exit without saving changes
    - `:q!`
  - To save and exit (**recommended**)
    - `:wq`

# Text editor – Vim (6)



# Text editor – Vim (7)

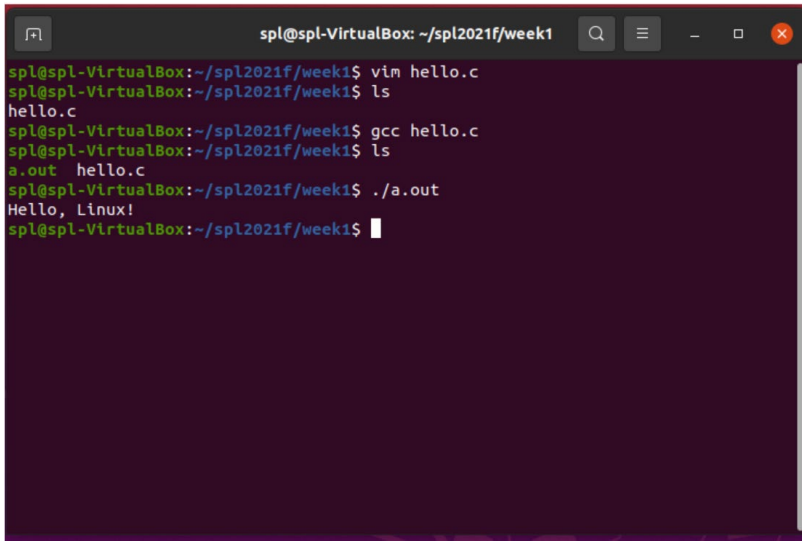


# Text editor – Vim (8)

- **For learning Vim:**
  - Vim Adventures (Game)
    - http://vim-adventures.com/
  - Vim Tutorial
    - http://www.openvim.com/tutorial.html
- **Repeat, repeat, and repeat**

# Lab Exercise

- **Write *hello.c* using C code with vim**
- ***hello.c* program prints out "Hello, Linux!"**
- **Submit only the screenshot via iCampus**
  - Filename: "STUDENT ID_w1.png"
- **Deadline: Next week (3/11 23:59)**



# Precautions

- **Using other virtual machine monitors (e.g. VMWare, UTM, etc.) or native Linux servers or computers is allowed.**
- **Using In-Ui-Ye-Ji cluster (인의예지 클러스터) for this course is strongly discouraged.**
- **You can use Ubuntu 22.04 or newer versions for this course. Using older versions is discouraged.**
- **Use other distros and WSL2 at your own risk.**
  - Programs compiled in these environments might behave differently, leading to grade deductions.
  - If you still decide to use WSL2, please use the newest version as the older versions lack some features needed for this course.

# Final remarks

- **Electronic attendance is used for this course.**
  - Either use the 성균관대학교 전자출결 (SAM) app or go to the reader to scan the Kingo-M QR code or tap your student ID card.
  - If you forget, you can request for attendance status change (출결변경신청).
- **Deadline for the exercises is midnight of the same day (Wed 23:59)**
- **Use the Q&A Board (문의게시판) to ask questions related to the course.**
  - You can also use the Q&A board during the exercise part of the class time to help others with the same question and to easily translate the answer to Korean or any other language.
  - Ask questions as early as possible, especially when the questions are related to the programming assignments (PAs).
- **Class ends at 21:45. If you have additional questions before the deadline, feel free to ask us via iCampus or e-mail.**

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# Any Questions?